

□ Real-Time Stock Market Analytics Pipeline using AWS

Services: Kinesis, Lambda, DynamoDB, S3, SNS, Glue & Athena

This document provides a detailed breakdown of the real-time stock analytics system, explaining the Lambda function code, the data pipeline workflow, and how each AWS service interacts with the others.

1. System Overview

The pipeline streams stock price data in real time using a Python producer script, processes it with AWS Lambda, stores raw and processed results in S3 & DynamoDB, triggers alerts using SNS, and enables historical analytics using Glue + Athena.

The workflow contains **five major components**:

Component	Purpose
Python Producer	Generates or fetches simulated stock market data
Kinesis Stream	Ingests real-time stock price events
AWS Lambda	Processes, transforms, and routes the data
DynamoDB / S3	Storage for structured (DDB) & raw (S3) data
SNS Alerts	Sends notifications on abnormal price spikes

2. Code Structure Overview

Your Lambda function is organized into four main logical sections:

Component	Purpose
Lambda Handler	Entry point coordinating the workflow
Data Extraction & Validation	Decodes Kinesis payloads
Database & Storage Layer	Saves raw + processed data
Anomaly Detection & Alerts	Sends notifications via SNS

3. Lambda Handler Function

Below is the exact Lambda handler used in your project:

```
def lambda_handler(event, context):

    for rec in event['Records']:
        payload = base64.b64decode(rec['kinesis']['data']).decode('utf-8')
        data = json.loads(payload)

        symbol = data["symbol"]
        price = data["price"]
        ts = data["timestamp"]

        # 1. Store RAW record in S3
        s3.put_object(
            Bucket=RAW_BUCKET,
            Key=f"{symbol}/{ts}.json",
            Body=json.dumps(data)
        )
```

```

# 2. Store processed record in DynamoDB
table.put_item(Item={
    "symbol": symbol,
    "timestamp": ts,
    "price": Decimal(str(price))
})

# 3. Simple anomaly alert
if price > 3500:
    sns.publish(
        TopicArn=SNS_TOPIC,
        Subject="Price Alert",
        Message=f"{symbol} crossed threshold: {price}"
    )

return {"status": "OK"}

```

What This Function Does

- Acts as the **central processor** for the stock data pipeline.
- Receives streaming records from **Kinesis**.
- Decodes Base64 payloads.
- Extracts important fields:
 - symbol
 - price
 - timestamp
- Sends data to:
 - **S3** → for raw analytics
 - **DynamoDB** → for structured queries
- Checks if the price exceeds a predefined threshold → triggers **SNS alerts**.
- Clean error handling ensures reliable processing.

4. S3 Raw Storage Functionality

What It Does

- Stores every incoming event as a JSON file.
- Organizes files by **symbol/** folders, improving query performance.
- Enables Athena to perform historical analysis on raw data.

Example File:

AAPL/2025-12-06T12:36:14.123456.json

Stored JSON:

```

{
  "symbol": "AAPL",
  "price": 149.76,
  "timestamp": "2025-12-06T12:36:14.123456"
}

```

Key Insights

- Raw data is immutable — valuable for audit trails.
 - Allows Glue Crawler to detect schema automatically.
 - S3 storage is extremely cheap → scalable for large datasets.
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5. DynamoDB Structured Storage Layer

What It Does

- Saves processed stock data for fast lookup.
- Uses **symbol + timestamp** as a composite key.
- Supports real-time dashboards or API querying.

Example DynamoDB Record:

```
{  
  "symbol": "AMZN",  
  "timestamp": "2025-12-06T12:36:31.599028",  
  "price": 3500.59  
}
```

Key Insights

- DynamoDB is ideal for real-time analytical workloads.
 - Each record is written with millisecond latency.
 - Perfect for building future dashboards.
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6. SNS Alert Notification

What It Does

- Sends an email alert when a stock exceeds a given threshold.
- Includes details such as symbol, price, and timestamp.

Example Email:

Price Alert: AMZN crossed threshold: 3503.46

Key Insights

- SNS ensures immediate notification of market anomalies.
 - You can extend the system to notify Slack, SMS, or webhook endpoints.
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7. Glue Crawler & Athena Integration

Glue Crawler

- Scans S3 raw data.
- Automatically infers schema.
- Creates a table in Glue Data Catalog.

Athena

Allows SQL queries such as:

```
SELECT symbol, AVG(price)  
FROM stock_raw_bobby  
GROUP BY symbol;
```

Key Insights

- Zero ETL pipeline — Athena reads data directly from S3.
 - Enables dashboards and trend reports.
-

8. CloudWatch Dashboard Monitoring

The dashboard includes:

Metric	Purpose
Lambda Invocations	Measures throughput
Lambda Errors	Detects processing failures
Lambda Duration	Performance tuning
Kinesis Incoming Records	Validates producer performance
DynamoDB Read/Write Units	Monitors scaling behavior
SNS Notifications	Tracks price anomalies

Key Insights

- Helps detect issues early.
- Provides live health overview of the entire data pipeline.

9. Environment Variables

Variable Name	Description
RAW_BUCKET	S3 bucket for raw JSON
TABLE_NAME	DynamoDB table storing processed data
SNS_TOPIC	SNS topic ARN for notifications

These allow updating configuration without modifying code.

10. Summary Table

Stage	Service	Purpose
1	Python Producer	Generates real-time stock data
2	Kinesis Stream	Ingests fast-moving data
3	AWS Lambda	Processes and routes data
4	S3	Stores raw event files
5	DynamoDB	Stores processed data
6	SNS	Sends alerts
7	Glue	Auto-detects schema
8	Athena	Performs historical analytics

Thank you for visiting